Wearable System Using Printed Interdigitated Capacitive Sensor for Monitoring Atopic Dermatitis in Patients

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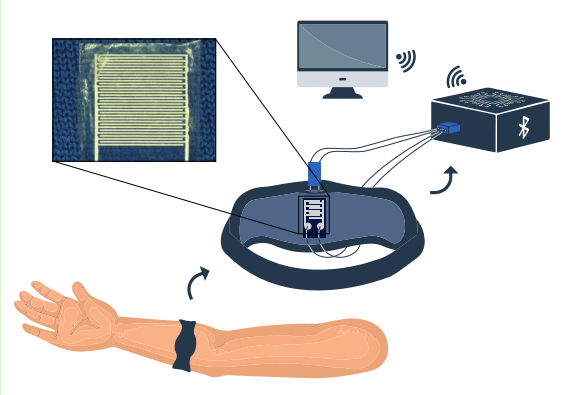
***Abstract*—Recent advances in sensor technology offer the potential to transform dermatology by enabling continuous monitoring and objective, data-driven assessment of skin conditions. This work presents a novel wearable device for non-invasive assessment of atopic dermatitis (AD) severity in patients. The device uses a bespoke interdigitated capacitor (IDC) sensor, sensitive only to biomarkers of AD, namely stratum corneum (SC) hydration. The sensor is integrated into a flexible textile armband and paired with a compact readout circuit, capable of transmitting real-time SC hydration data via a custom graphical user interface (GUI). The device exhibited excellent measurement repeatability and stability under different environmental conditions. It was tested on 13 patients with the condition and demonstrated strong correlation with the standard clinical assessment tools such as the Corneometer ($r = 0.595$, $p < 0.05$). The e-textile IDC sensor identified a difference of 3–5 pF between skin with symptoms of the condition compared to skin without, while showing significantly less variability compared to the Corneometer. The improved stability and accuracy, combined with the conformal form-factor and ability to perform continuous measurements make the e-textile IDC sensor a much better candidate for at-home monitoring of AD in patients, compared to the current standard tools.**



***Index Terms*— Atopic dermatitis (AD), e-textile, interdigitated capacitor, medical, remote monitoring, wearable sensors.**

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I. Introduction

WEARABLE sensors and remote sensing technologies have become essential for advancing personalized healthcare monitoring at home [1]. These innovations have significantly impacted various medical fields, enabling continuous and real-time monitoring of patients' health metrics. For example, wearable ECG devices have transformed remote cardiac care by tracking multiple patients at a time, and similar devices facilitate data collection of patients with neurological disorders [2]. Providing specialized medical care and guidance through mobile Internet of Things (IoT) platforms is especially beneficial for people in rural or remote locations or for patients who are unable to leave their homes. Despite these advances, medical fields like dermatology and the monitoring of common skin disorders have not fully benefited from these innovations [3], [4].

Diagnosis and treatment of conditions like atopic dermatitis (AD) still depends on in-person inspections by a doctor. There is no accepted empirical measurement to objectively determine the severity of AD. Instead, evaluations rely mostly on scoring systems estimated by visual observa-

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(a) (b)

Fig. 1. Photographs of AD-affected areas in skin. (a) Lesions on the back and neck. (b) Lesions on the wrist.

tion done by a dermatologist [5]. Furthermore, there is no way for patients to continuously monitor their treatment—everything is done through invasive medical tests on site. Often, patients have to revisit the clinic and get reassessed every 4–5 months to assess the efficacy of the treatment. This introduces a significant risk of misdiagnosis or incorrect choice of therapeutic procedure [6]. There are commercial instruments that attempt to solve this, but their high cost has limited their use to clinical and research environments. The Corneometer[1] and the Tewameter[1] by Courage+Khazaka (Courage+Khazaka GmbH, Cologne, Germany) are the current gold standards used in the literature when assessing AD severity, but they are not optimized exclusively for AD monitoring and are not accepted by any diagnostic criteria [7]. Thus, there is a need for a device that can remotely and accurately quantify the severity of AD and monitor its treatment overextended periods. The device should be affordable, portable, and noninvasive or wearable, to allow for seamless integration into the daily routines of patients while providing clinicians with useful data for long-term remote monitoring.

AD is the most common inflammatory skin condition worldwide, as one in ten people will develop it during their lifetime [8]. The disease comprises dysfunction of the skin barrier, primarily in the outermost layer of the skin, including the stratum corneum (SC). This barrier protects from external influences like allergens and viruses and prevents excessive dehydration. Thus, areas of the skin affected by AD are red and inflamed, easily irritated by environmental triggers, and are scaly and dry. These changes are associated with increased loss of water through the SC and the epidermis (EP) [9]. The rate at which the skin is dehydrated is called the transepidermal water loss (TEWL) and is an indicator for the disrupted barrier function. The dryness of the SC and EP is also a biomarker of AD severity and can be monitored through measurement of the dielectric properties of these layers. Fig. 1 presents the photographs of the affected areas of skin with AD.

In a previous publication, the authors have addressed the lack of empirical measurements for assessment of AD severity by proposing an Interdigitated capacitor (IDC) sensor tailored for the application [5]. The design of the proposed IDC is shown to be sensitive to dielectric changes only in the SC and EP layers, allowing for varying states of AD severity to be accurately determined [5]. The sensor was characterized and tested on skin phantoms, which replicate skin with the condition, and its behavior was found to be stable under varying environmental conditions [5]. The IDC sensor has also been printed onto a textile armband to showcase its utility as a wearable device monitoring the hydration of the SC of healthy volunteers [10].

Here, the authors report the integration of the IDC sensor armband with a bespoke readout circuit, creating a mobile e-textile monitoring system for AD severity, with data collated remotely by Bluetooth. The system is fully portable and would allow patients to monitor their condition outside of the clinical environment. The wearable form-factor allows for longitudinal measurements to be performed, introducing a new dimension that surpasses the current gold standard testing. The device was tested on 13 patients with the condition, aged 18–70, and the results reveal that the sensor can successfully distinguish between affected areas (lesional) and unaffected areas (nonlesional) of AD skin, offering a digital measurement of AD severity.

II. Methodology

A. IDC Sensor Operation

The IDC sensor operates like a capacitor, where the measured capacitance is directly proportional to the relative permittivity of the medium in contact with the capacitor, which in this case is the outermost skin layers [6]. Water molecules significantly affect the electrical properties of these layers, by creating ionic pathways through the introduction of more dipoles, capable of storing electrical energy [6]. Therefore, high water content results in an increased dielectric permittivity and in the case of AD, the decreased water concentration of the SC and EP layers would translate into a lower value for the relative permittivity of these layers. The lower permittivity can be recorded through the measurement of capacitance, as they are directly proportional. This establishes a direct relation between measured capacitance and the dehydration of the SC and EP layers, for an IDC that is sensitive exclusively in these layers [6]. The IDC features a repeated pattern of intertwined electrodes of opposing polarities, separated by a specific distance, known as the gap (G). An alternating voltage signal is imposed on the electrodes to generate an electric field that penetrates the skin. To achieve sensitive measurements of a desired skin layer, the gap between the electrodes is altered with smaller gaps resulting in shallower penetration depth of the electric field lines and therefore increased sensitivity to changes in the relative permittivity of these layers [5]. The addition of an encapsulation layer also affects the penetration depth, which will be discussed below. Fig. 2(a) shows a planar view of the IDC sensor used with its dimensions. The IDC's sensing area is 0.384 cm^2. Fig. 2(b) presents a cross section of the electrode against the skin.

In previous works, the authors have simulated and empirically tested IDC designs of varying gap distance and resolution, to conclude a pattern best optimized for measuring changes in the hydration of SC and EP layers [5], [10], [11]. The IDC electrodes were tested on bilayer skin phantoms with

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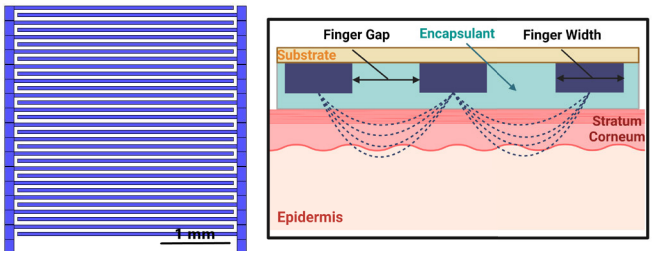

(a) (b)

Fig. 2. (a) Planar view of the IDC sensor, featuring a repeated pattern of 32 interdigitated electrodes, called fingers, with a 100 μm width and a 100 μm separation between the fingers called the gap. The length of each finger is 6 mm. (b) Side view of the IDC electrode with notation on the different components and terminology, along with illustrative lines indicating the effect of the electric field.

changing dielectric properties, simulating different severity levels of AD. It was discovered that an IDC with a bespoke pattern of 32 electrodes of 100 μm width separated by a 100 μm gap yielded the greatest change in capacitance per change in the severity of AD in the skin phantom [5]. This work was then continued in a separate publication, where electrodes separated by a gap of 50 μm were investigated, along with varying encapsulation layer thickness. It was concluded that with a thinner coated encapsulation of less than 10 μm, the gap of 50 μm is more sensitive to changes in the SC, but with a thicker encapsulation of 50 μm, the gap of 100 μm is better [12]. In this work, the second arrangement is chosen (encapsulation of 50 μm, electrode separation of 100 μm), because it is easily scalable, as the sensors and encapsulation are easily printed and laminated onto a wearable strap band to be tested on healthy volunteers. The results confirmed the initial testing by successfully distinguishing between normal and moisturized skin of volunteers and could also monitor the efficacy of treatment creams after application [10]. In a separate publication, a novel, layer-separated FEA model of the skin with ADs was presented and used to determine that the most optimal frequency for the excitation signal that drives the IDC is in the range of 1 kHz–1 MHz [11]. The culmination of this research is presented here, where the IDC sensor is fabricated onto an e-textile wearable device for noninvasive assessment of the severity of AD in patients.

B. Device Fabrication

1) IDC Electrodes: The bespoke IDC pattern was printed onto TPU transfer film using a method called reverse offset printing, developed by Dai et al. [12] demonstrated that the method achieves higher resolution and better reproducibility for AD monitoring than screen printing. In this method, a PDMS-covered roller was coated uniformly with Dupont 5000 silver conductor paste (Dupont, Bristol, U.K.) using a doctor-blade coater. The ink was transferred to a patterned cliché with high surface energy, leaving a negative image on the roller. The remaining ink is then deposited onto the TPU substrate to create the desired pattern [13]. The substrate was Electrum Stretch HC with a thickness of 80 μm (Policrom Screens S.p.A., Carvico, Italy). After printing, the ink was cured at 120 °C for 15 min in a carbolite box oven. The

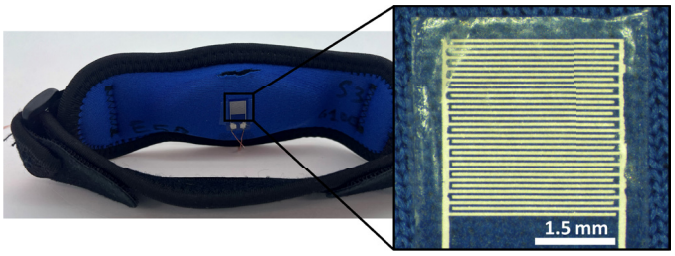

Fig. 3. Photograph of the e-textile IDC sensor for monitoring AD in patients and a micrograph of just the IDC sensing element—an array of 32 interdigitated electrodes with a 100 μm width and a 100 μm separation between them.

finished print was then laminated onto the skin-facing side of the elbow strap band (Ionocore, U.K.) using a hot press at 160 °C for 1 min. The strap band was selected as a substrate for the IDC print because it has a cushion on its skin-facing side, capable of applying constant pressure on the skin. This achieves a better contact between the sensor and the skin, partially eliminating air gaps from forming. To electrically isolate the sensor from the skin and to improve the conformal contact, the IDC electrodes were encapsulated with another layer of TPU film with a thickness of 50 μm—Platilon U073 TPU film (Covestro Ltd., Cheadle Hulme, U.K.). The encapsulation layer was proved in a previous publication not to change capacitance sensitivity [12]. Fig. 3 shows a photograph of a finished e-textile strap band with an inset of the printed IDC on the inner part.

2) Capacitance Measurement Device: From FEA simulations and tests on skin replicas in previous publications, it was discovered that the measurement range of a sensor able to distinguish between lesional and nonlesional skin of patients is in the tens to hundreds of picofarads. The range is wide due to the ideality of simulation, which does not accurately reflect the uneven surface of the skin, and the presence of air gaps between the skin and sensor surface. Nevertheless, a measurement circuit suitable for use in this range is needed to convert the capacitance values to a digital signal locally and avoid the usage of standard benchtop multimeters and impedance analyzers. The digital data can then be collected by a microcontroller unit (MCU) and transmitted wirelessly to a host computer for further processing.

The FDC2214 capacitance to digital converter (CDC) chip by Texas Instruments (Texas Instruments Ltd., Manchester, U.K.) was chosen as the analog front end measuring the IDC's capacitance due to its high resolution, serial data transmission, and low power consumption—around 0.3 mA per channel at 3.3 V with a 1 MHz measurement frequency [14]. The FDC2214 features four channels, each with 28-bit resolution, sampling the resonating frequency of an external LC tank. Only one channel was used in this application, but the extra channels make the chip an attractive option for scaling the device to cover larger areas. The high resolution allows the CDC to detect capacitance changes in the order of femtofarads, which is more than enough for the given application. The values of the external inductor and capacitor dictate the excitation frequency of the electric field emitted onto the skin, based on (1). When an unknown capacitance is added to the circuit, the resonance frequency shifts, and the FDC2214

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measures the new frequency

$$f = \frac{1}{2\pi\sqrt{LC}}. \tag{1}$$

Based on the digitized frequency value, the value of the unknown capacitance can be calculated, which in this case is the capacitance of the IDC sensor in contact with the skin, by taking the inverse of (1) and subtracting the capacitance of the LC tank – exemplified in the following equation:

$$C_{\text{sensor}} = \frac{1}{L\,(2\pi * f_{\text{sensor}})^2} - C. \tag{2}$$

The LC tank values were selected to achieve an excitation frequency of 1 MHz, by using a 470 μH inductor and a 47 pF capacitor. To ensure stability of the LC circuit, a high Q-factor is critical, and the selected components have Q values of 200 for the inductor and 1000 for the capacitor, resulting in a combined Q-factor of 167, well above the recommended ($Q > 50$) by Texas Instruments for the FDC2214 [15], [16]. The overall tolerance shift in the excitation frequency is ±6%, which still fits within the high sensitivity range, discovered during previous FEA simulations, and is also the same frequency used by the Corneometer, ensuring a comparable basis between the two devices. Based on the LC tank and the measurement range, a drive current is selected. The output voltage of the LC tank has to be between 1.2 and 1.8 V, as per the FDC2214 datasheet. Measurements confirmed that the most suitable value for the drive current is 3 mA, and that value was used throughout the data collection [14]. The LC tank is highly temperature-stable (±30 ppm/ °C). These factors collectively ensure sufficient resonance stability for wearable applications.

To sample the frequency data from the CDC an XIAO nRF52840 module by Seeed Studio was used (Seeed Studio, Düsseldorf, Germany) [17]. The MCU polls the latest value from the FDC2214 every 1000 ms, resulting in a system sampling rate of 1 Hz. A custom PCB with all the necessary components to drive the FDC2214 was created, and the XIAO module was soldered to it. The module has the nRF52840 SoC by Nordic Semiconductor and it features Bluetooth low-energy (BLE) connectivity with a matched on-chip antenna, I2C serial transmission at 3.3 V operation—suitable for interfacing to the FDC2214 [18]. A 3.7-V 620-mAh LiPo battery was used to power the XIAO module, which has a built-in 3.3 V voltage regulator and a power management circuit for charging the LiPo battery back through a USB port. Datasheets for the XIAO nRF52840 and FDC2214 IC indicate a current consumption of approximately 25 mA, translating to 24.8 h of continuous usage [14], [17]. However, the device was kept fully charged in between patient tests, to ensure no power failure during the procedure, so this limit was never investigated.

Pressure is an important parameter to control because if too much pressure is applied, the layers of the skin are compressed, which alters the depth of the electric field and affects the results. So, to ensure consistent and repeatable pressure application between readings, a FSR06 thin-film pressure sensor (Ohmite, Warrenville, IL, USA) was fit in the cushion of the strap-band, directly underneath the IDC sensor. Fig. 4 shows a photograph of the pressure sensor and its position behind the IDC sensor. The pressure sensor acts as a variable resistor with one leg connected to power and the other connected to an ADC input pin of the MCU. Increasing the pressure decreases the resistance of the FSR06 and thus the ADC reading of the voltage increases. Through this technique, pressure was measured constantly, and thus, the capacitance reading from the FDC2214 chip was transmitted only when the pressure was within a certain predefined range. This range was empirically tested during trials to ensure proper conformal contact of the sensor on the skin without patient discomfort and set constant for the duration of the study.

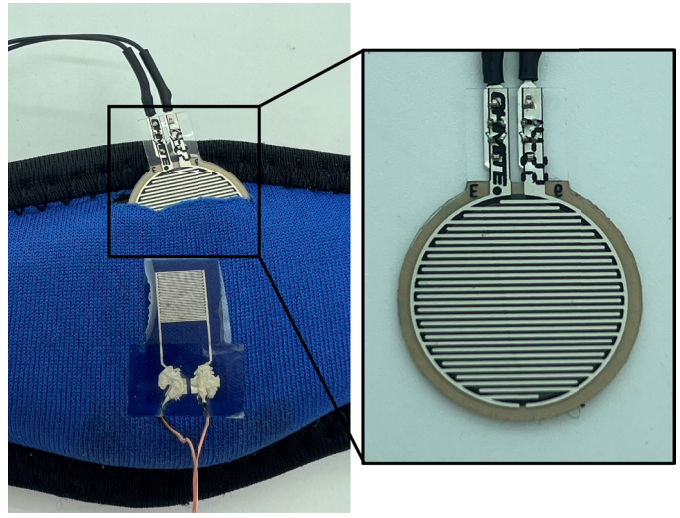

Fig. 4. Photographs of the thin-film pressure sensor used to control the IDC sampling, fit in a cutout on the side of the textile armband, directly underneath the printed IDC sensor.

After building the finished device, the excitation frequency was verified with an oscilloscope. It was discovered that the actual frequency is around 0.95 MHz, attributed to parasitic capacitance in the PCB tracks, calculated to be 12 pF, and the capacitance of the IDC sensor itself, which is approximately 1 pF. This frequency is still within the high sensitivity range, so there were no alterations to the hardware. Possible mitigation options for future design would be to use shorter traces to sensor leads or an entire ground shield or ground plane directly underneath the LC circuit.

3) Device Operation and GUI for Data Visualization: To complement the IDC device and to extract the data wirelessly from it, a custom graphical user interface (GUI) app was developed in Python 3.12. The GUI allowed a user to connect via BLE to the IDC device, visualize the data in a table, and save the recorded data as a. csv file for further processing. The entire process of operation of the IDC system (device + app) is simplified with a flowchart in Fig. 5. When the IDC device is switched on, it runs a setup sequence that starts advertising the BLE service and initiates the FDC2214 to configure the desired channel and use an external 40-MHz crystal oscillator as a clock source. Then, the IDC device waits for a client to subscribe to the BLE service. On the GUI app, the user can press a button to connect to the device, but no data is

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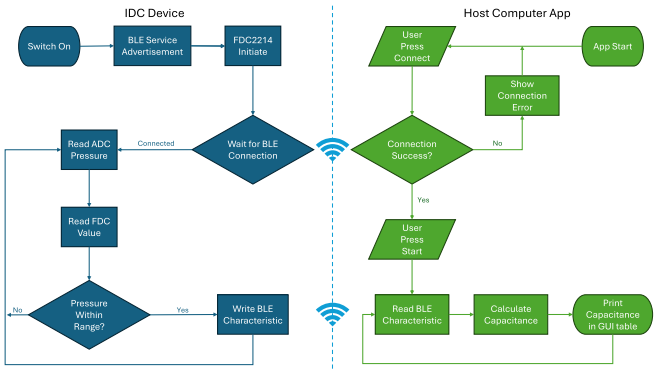

Fig. 5. Flowchart describing the capacitance measurement and data transmission of the IDC wearable device. All of the processes performed by the IDC device are colored in blue, whereas the tasks undertaken by the GUI application are colored in green.

transmitted yet. If the connection is successful, the IDC device starts the data collection sequence. In this sequence, which repeats every second as dictated by a delay from the MCU, ADC pressure data and FDC2214 frequency readings are sampled (payload < 20 B). With default connection parameters, the end-to-end latency is sub-second and negligible relative to the 1 s sampling period, and no missing samples were observed during typical short acquisitions. If the pressure reading is within a predefined range, the data are written to the BLE characteristic register and advertised for a client to read. On the GUI app, the user can press the start button, which subscribes the host computer to the notifications from the BLE advertisement, which are updated every second following the pressure-FDC reading sequence. Once a package of FDC frequency readings is read, the capacitance is calculated and printed on a table in the GUI. Afterward, the user can choose to stop the data transmission with a stop button functionality (not shown in the flowchart), unsubscribing from the BLE service, and save the data in the GUI table as a. csv format or clear the table with a save data and clear data buttons, respectively.

Tests were set up to run for 30 seconds of continuous use and collect a continuous data stream at the desired pressure range. After the 30 s have passed, a special notification prompts the user to remove the sensor, stop the data collection, and save the data before continuing to the next test. Further trials will explore longer wear tests.

C. Data Collection

Before testing the IDC device on patients, a sensor characterization study was performed. The repeatability and stability of the device's measurements were investigated across several days using a material with a known dielectric permittivity to act as a calibration. Then, the IDC capacitance readings were recorded while varying the environmental humidity, to determine if the room conditions have an effect on the sensor.

In the patient trials, 13 patients with AD aged 18–70, both male and female, participated in the study. The patients were recruited from the dermatological clinic, part of the University Hospital of Southampton. The study was done in conjunction with the NOMAD study, ethics approval: NRES 330915. Their skin was assessed by a dermatologist following accepted diagnostic criteria and specific areas of lesional and nonlesional skin on the arms of the patients were identified by the dermatologist. Lesional skin were areas with acute symptoms of AD, usually inflamed, red, and dry. Nonlesional skin were areas where there were no visible symptoms, and the skin appeared similar to clinically healthy skin. In some patients, almost the entire body was covered in lesions, so the least affected skin area was selected as nonlesional.

The e-textile device was located on lesional and nonlesional areas of the skin along various places of the patients' arms. The most common locations were the volar forearm and the anterior bicep. Several measurements were taken on each location to investigate sensor repeatability, with 1 min delay between measurements. The Corneometer and the Tewameter were used alongside the IDC device to correlate the readings to the current gold standard in skin hydration assessment. Ten readings with each of the gold standard devices were taken from the same lesional and nonlesional area on the skin, where the capacitance was sampled. Readings were taken right after the corresponding reading with the IDC device finished. The Tewameter was used on only seven of the 13 patients.

III. Sensor Device Characterization

A. Repeatability Test With a Dielectric Calibrating Weight

To validate the correct operation of the device, the stability of the readings across different days was tested before progressing with patient testing. A 3-D-printed weight made of PLA material with a known and constant dielectric permittivity was used to test both the capacitance and the pressure sensor's outputs. In our previous publication, it was shown that under stable and repeatable conditions, the IDC's measured capacitance does not fluctuate by more than 2 pF. The goal in this test is to confirm the result and verify that the developed system stays within that confidence interval and does not introduce too much noise into the measurement.

The calibration weight was 5 N and it was positioned on top of the skin-facing side of the strap band, in contact with the printed IDC sensor. The testing setup is presented in Fig. 5. The weight was kept for 30 s, to replicate the setup during patient tests, and for each day, the test was repeated four times, and the results were averaged. The relative permittivity of PLA is 3, and based on the dimensions of the IDC sensor (distance between electrodes and effective area), it is calculated that the capacitance will increase by 4 pF when the calibration weight is placed in contact with the sensor. Tests were performed under standard room temperature of 22 °C and relative humidity of 38%.

The results, presented in Fig. 6, reveal the experimental setup, the stability of the capacitance measurements, and the pressure sensors against time. It can also be observed that the readings are repeatable across different days. The capacitance of the sensor increased by about 1 pF with the addition of the calibration weight—from 38 pF when in contact with air, to 39 pF. This is less than the calculated value, but within the reasonable range. As the weight's surface is not fully flat, it is not possible to completely contact the IDC sensor, and thus the addition of air in the sensor–PLA interface reduces the

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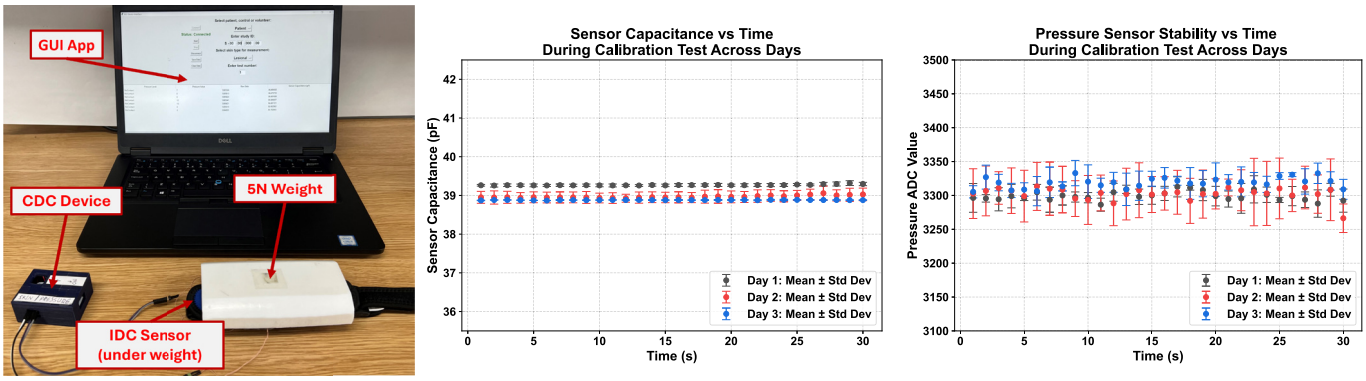

(a) (b) (c)

Fig. 6. (a) Photograph of the test setup used for the calibration testing. The IDC sensor is positioned under the 5 N Weight and is connected to the CDC device that converts the capacitance to a digital signal. (b) Sensor capacitance against time during the calibration test across several days. (c) Pressure sensor ADC readings against time during the calibration test across several days.

measured capacitance. A slight trend of decreasing capacitance on each consecutive day is observed, but the decrease is insignificant as it is less than 0.5 pF, or less than 2% of the total capacitance value, within the margin of error. The ADC value of the pressure reading remained around 3300 for the calibration weight across all 3 days. Without the weight, the ADC value read was 0 for all 3 days. The calibration force of 5 N, used to achieve an ADC value of 3300, equates to approximately 130.2 kPa of pressure across the surface of the sensor.

B. Relative Humidity Test in a Climate Controlled Chamber

The e-textile IDC sensor was also tested against changing humidity levels in the environment, to determine the effect of relative humidity on the sensor's performance. Temperature and mechanical bending were previously found not to affect sensor performance [5]. The IDC printed sensor was placed in a WKL100 climate chamber (Weiss Technik, Loughborough, U.K.), and its capacitance was measured continuously with the CDC, which was situated outside of the chamber. The following humidity profile was programmed into the chamber: 5 min at 40% RH; 20 min gradual increase to 90% RH with a step of 2.5% RH per min; 10 min at 90% RH; 20 min decrease back to 40% RH; 5 min at 40% RH—for a total time of 1 h. To measure the actual humidity, a TE-03 TH temperature and humidity data logging probe was also inserted into the climate chamber (Thermo Electric Company, West Chester, USA). The result of the test is presented in Fig. 7. The capacitance of the IDC sensor changes with humidity, but for the most part, the effect is minimal, less than 0.5 pF change when varying between 40% RH and 80% RH. Only when RH rises above 80%, a capacitance change of 1.5 pF is observed. The two spikes in capacitance readings at the beginning and at the end of the 90% RH programmed time can be attributed to a condensation up of water molecules on the surface of the IDC. The chamber maintains humidity by constantly switching a ventilation fan on and off, and the sudden trough after the first spike is caused by the evaporation of the water droplets due to the ventilation fan turning on. The uneven profile of the actual RH measured reveals many fluctuations in humidity, indicating frequent ventilation events.

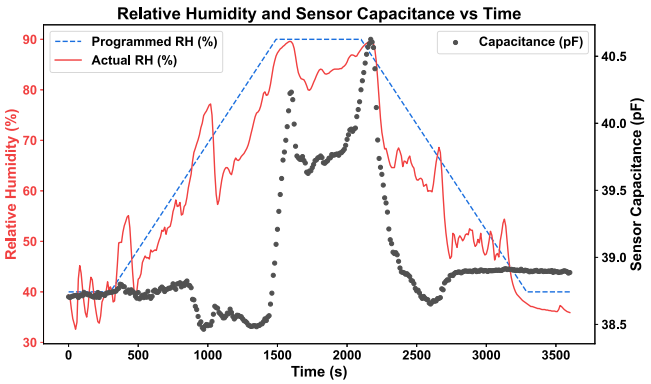

Fig. 7. Change in capacitance of the IDC sensor over time for a varying change in the relative humidity of the environment.

The two troughs of capacitance before and after the 90% RH range are also due to the ventilating fan, as confirmed by the behavior of the actual RH measured in red. These troughs decrease the capacitance by about 0.4 pF, which is within the margin limit. At the end of the humidity test, the capacitance readings return to almost the same value as the start—within 0.2 pF. Overall, the IDC shows impressive stability under extreme environmental humidity conditions, as the change in capacitance is negligible compared to the change arising from a patient's skin.

IV. Results and Discussion

As described in the methodology, the IDC sensor was used to sample a 30-s continuous measurement of the integrity of the SC barrier for each of the 13 patients. The pressure used in the characterization tests of 3300 ADC counts or 130 kPa was too high and caused discomfort in the patients; thus, it was reduced to 2100 ADC counts or 65 kPa. This pressure value was maintained throughout the entire patient study. Discrete measurements with the Corneometer and Tewameter were then collected. Fig. 8(a) presents the IDC capacitance continuous data for one of the patients (number 13), along with the averaged single-point-in-time measurements for the corneometry and TEWL. In Fig. 8(b), a photograph of the skin of patient 13 is shown along with labels highlighting the lesional and nonlesional skin areas selected for this patient.

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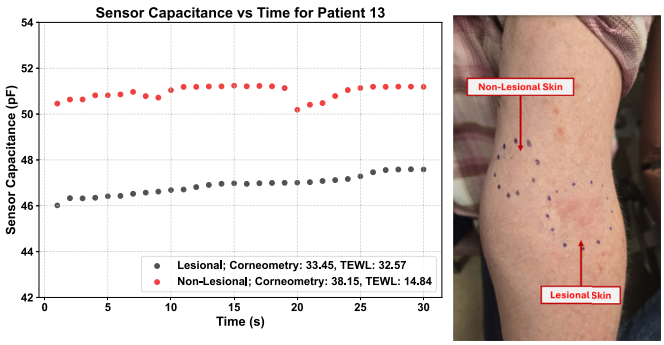

(a) (b)

Fig. 8. (a) IDC sensor capacitance against time for an individual patient with AD on lesional and nonlesional areas on the skin. The legend shows the corresponding corneometry and TEWL values. (b) Photograph of lesional and nonlesional areas of the skin of the right arm of patient 13.

The two spots selected were both on the patient's right arm.

The visual difference between the two skin areas is apparent, and the capacitance readings strongly align with that. In this case, TEWL readings for the lesional skin are in the high 38 g/m^{2}h, hinting toward greater loss of barrier integrity and therefore higher severity of AD in skin.

The capacitance, corneometry, and TEWL data for each of the patients are aggregated and presented in Fig. 9. The range of capacitance readings is between 40 and 55 pF across all patients, with lesional skin readings having an average of 46.71 pF and nonlesional skin readings an average of 49.25 pF. This reveals a considerable measurable difference between lesional and nonlesional skin of patients, which is also visible across all patient readings in Fig. 9(a). In patients 4 and 13, this difference is the greatest, with corresponding delta values of approximately 6 pF and 4 pF. Patient 1 featured the lowest capacitance values of 40–42 pF, indicating that this condition was the most severe from the perspective of the IDC sensor. This is supported by the highest measured TEWL values (35–50 g/m^{2}h) and low corneometry values (16–26 A.U.) for patient 1. TEWL is a measure of the water loss in skin, and corneometry is a marker of skin hydration. Therefore, lesional skin with AD should exhibit high TEWL and low corneometry, which is the case here and is in line with the low IDC capacitance of patient 1. The values suggest that the condition has progressed so much in this patient, it also affects the visually nonflaky regions of the skin, thereby explaining the low capacitance and high TEWL values for nonlesional skin. The second-lowest capacitance readings are of patient 12, where there is also a small capacitance difference between the lesional and nonlesional skin, strengthening the claim that in very severe cases of AD, the barrier integrity in all areas of the skin is damaged.

For the Corneometer and Tewameter, the data are presented in Fig. 9(b) and (c), and it is the median value of ten iterative measurements for each patient. Median values were used instead of the arithmetic mean because it is a more accurate representation of the data extracted from the instrument, as it diminishes the effect of outliers from improper instrument application. Calculating the mean across all patients for the lesional skin gives a corneometry value of 21.9 A.U. and a TEWL of 33.86 g/m^{2}h, and for the nonlesional skin −31 A.U. and 19.92 g/m^{2}h. Both gold standard instruments reveal a difference between the two skin types, which is the same as with the novel textile IDC sensor, thereby confirming the latter's validity as an instrument toward identifying SC barrier integrity in patients with AD. This also highlights the positive correlation between the textile IDC sensor and the Corneometer and the inverse correlation of the former two to the Tewameter. Pearson's correlation coefficient for the combined lesional and nonlesional data from the IDC and Corneometer was found to be 0.595, with a p-value of 0.001, indicating a statistically significant ($p < 0.05$) strong positive correlation. Fig. 9(d) reveals the relationship. Pearson's correlation was not calculated for the TEWL readings, as there were too few to achieve statistical significance.

From Fig. 9(b) it is evident that the Corneometer readings feature more variability than the IDC sensor's readings, attributed to the wider standard deviation error bars. In the case of patient 13, the lesional and nonlesional error bars overlap, signaling issues with repeatability. This could be caused by an issue with calibrating the device. Another reason for the variability is the amount of applied pressure. The Corneometer's probe features a spring-loaded sensor head that depresses once in contact with the skin to ensure repeatable pressure application. Although reliable, this technique is prone to human error because even with the spring, a user can accidentally exert more pressure than needed, which alters the results. Furthermore, the rigid sensor head of the Corneometer does not conform to the curves of the skin.

The proposed e-textile IDC sensor solves all those issues by measuring direct capacitance values, monitoring the pressure, and its flexible substrate establishes conformal contact to the skin. The e-textile IDC device surpasses the capabilities of the commercial Corneometer, because it can provide longitudinal readings of the skin's severity, rather than just single-point measurements. The stability of the IDC sensor across time is evident with the small error bars in Fig. 9(a). In some cases, a slight capacitance drift is noticed in the first 5–10 s of putting on the device, and that is attributed to the skin conforming around the sensor. One limiting factor for the IDC sensor is occlusion, as the encapsulation and IDC electrodes are not water permeable, so condensation would build up on the skin–sensor interface. For the short test of 30 s, this effect is negligible. To prove the repeatability of the sensor readings, the longitudinal tests were conducted several times, with a 2-min wait between tests, on patient 8 as shown in Fig. 10 using the e-textile IDC. The maximum deviation between repeated IDC sensor readings was around 0.4 pF. This is much lower than the difference between lesional and nonlesional skin, so unlike the Corneometer, the IDC sensor does not obscure meaningful differences due to measurement variability. In contrast, the Corneometer showed greater inconsistency, with variations between repeated measurements sometimes exceeding the actual difference between skin types, making it less reliable for distinguishing AD severity.

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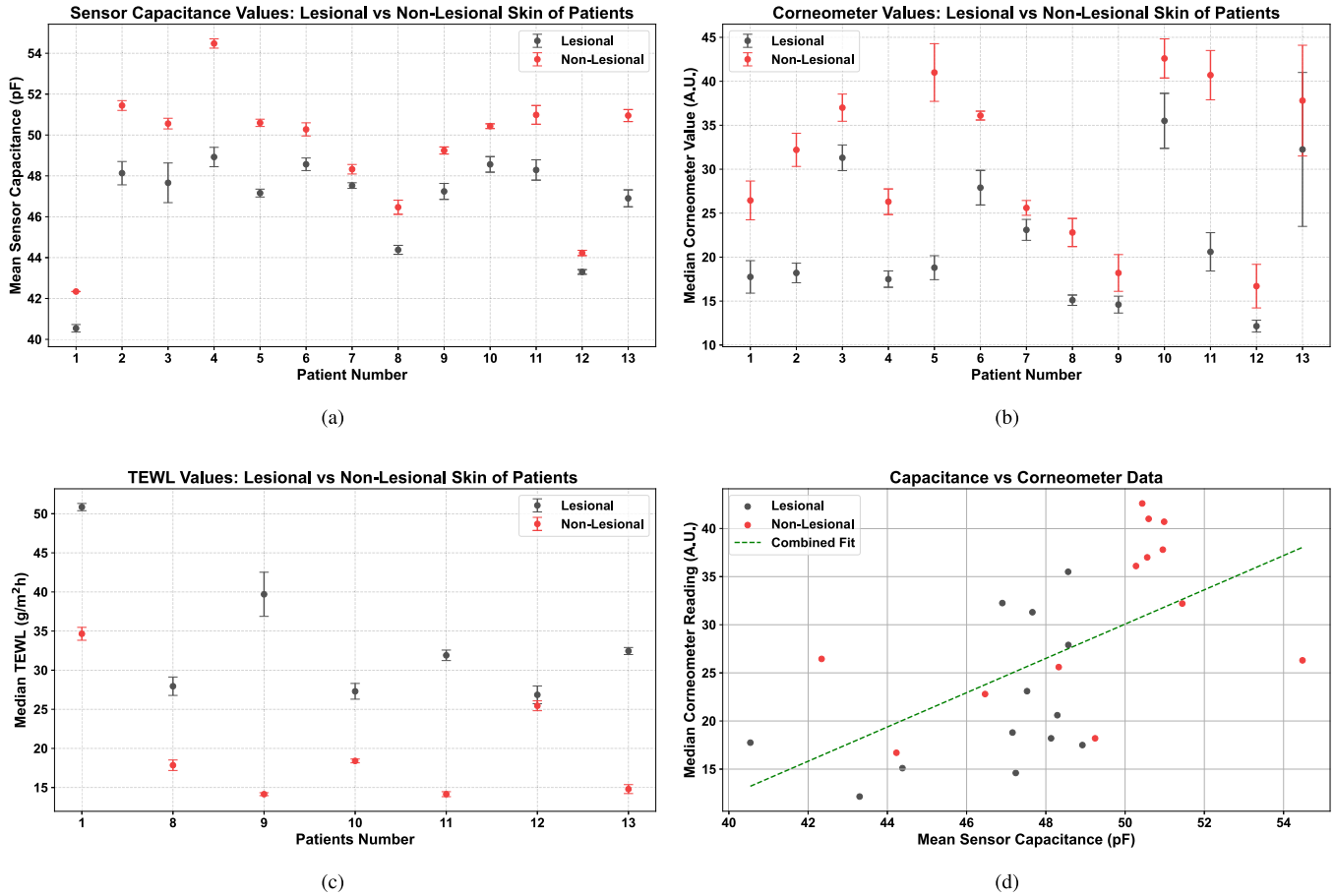

Fig. 9. Results from skin barrier integrity tests on lesional and nonlesional skin of the 13 patients with AD. Error bars indicate standard deviation. (a) Mean sensor capacitance values using novel textile IDC sensor. (b) Median Corneometer values. (c) Median TEWL values. (d) Correlation plot between textile IDC capacitance and Corneometer data, with a combined (lesional + nonlesional) best fit line plotted in green, showing the positive correlation between the two measurands.

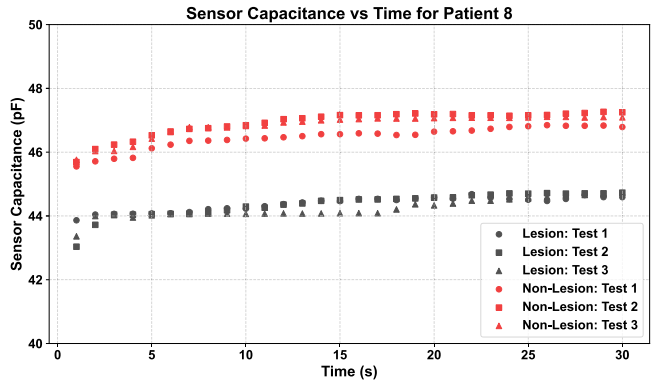

Fig. 10. Repeated IDC sensor capacitance against time readings for an individual patient. Three tests were performed for each skin area—lesional and nonlesional.

V. Conclusion

In this publication, a novel e-textile wearable sensor for assessing the severity of patients with AD in vivo is presented. The sensing element is optimized for screening biomarkers of the disease, and in this case, the hydration of the outermost layers of the skin, the SC and EP. Previous work into this sensor proved its accuracy and high sensitivity when tested on skin phantoms, and this work builds up on this by integrating the sensor into a portable device that is tested on patients with the condition. The portable device featured a compact read-out circuit that transmits the data wirelessly via BLE over to a host computer running a custom-built app. It showed repeatability in its readings across several days during the same test conditions, and stability when subjected to extreme humidity environments. The sensor successfully managed to distinguish between lesional and nonlesional skin in all the patients with AD that participated in this study, giving 100% sensitivity and specificity for individual patients. A single population-wide threshold could not be defined as the IDC sensor is most valuable for tracking intraindividual biophysical change rather than serving as a stand-alone diagnostic tool.

The readings showed a strong positive correlation with a gold standard in skin hydration assessments—the Corneometer. The IDC sensor proved to be better than the Corneometer because it featured less variability in the results and correctly identified high-severity patients alongside TEWL readings, which the Corneometer did not manage to do. Furthermore, the e-textile IDC device's continuous operation and flexible substrate allow longitudinal and improved conformal skin hydration measurements. This study revealed the importance of empirical assessment alongside visual observation when diagnosing AD and administering treatment. Future work will see the implementation of data analytics methods and machine

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learning for correlating the data and for providing new insights into better understanding the disease. Further work will be undertaken to scale the IDC sensor to cover wider areas of the skin and miniaturizing the electronics and the footprint of the read-out circuit to be fully integrated onto a wearable form factor. A textile pressure sensor will be implemented to create a seamless textile garment that can be worn by patients continuously. Patient-centric studies will be conducted to gauge the feedback from wearing the device and further improve the wearable IDC. The goal is to produce an easy-to-use, non-invasive device, that can be integrated with patient routines at home and can provide real-time data on their condition.

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VI. Data Availability

Data published in this article are available from the University of Southampton Repository at 10.5258/SOTON/D3507.

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